GLOBE SCOTT MOTORS · FREE ENGINEERING RESOURCE

Position Control Motor Selection Worksheet

For OEM machine designers specifying a motor, feedback device, gearing and brake for a positioning axis

Complete the sections in order. Sections 1 to 3 define what the axis must achieve and where errors come from; Sections 4 and 5 check that the chosen motor can move and hold the load. Send the completed sheet with your enquiry.

Section 1 — Define the requirement at the load

- ☐ Stroke or angle of travel, and whether the axis needs two fixed positions or continuously variable positions.
- ☐ Tolerance at the load in mm or degrees, stated separately for repeatability and for absolute accuracy.
- ☐ Direction of approach to each position: always the same side, or both directions.
- ☐ Move time, dwell time and number of moves per hour.
- ☐ Load mass or inertia, process force at the end of stroke, and axis orientation (horizontal, vertical or inclined).
- ☐ Required behaviour on power loss and E-stop: hold, free-wheel or move to a safe position.

Section 2 — Feedback selection

- ☐ Counts per output revolution = PPR × 4 × gear ratio.
- ☐ Resolution at the load = 360° / counts (rotary) or screw lead / counts (linear).
- ☐ Make the resolution 4 to 10 times finer than the tolerance.
- ☐ Choose absolute feedback if re-homing after power loss is not acceptable.
- ☐ Hall counting (6 × pole pairs states per motor revolution) suits only coarse positions through a gearbox.
- ☐ Decide whether the encoder sits on the motor or on the load; a motor encoder does not see backlash.

Section 3 — Mechanical error budget

- ☐ Output backlash in degrees, converted to lost motion at radius r: r × backlash in radians.
- ☐ Torsional wind-up under load: torque / torsional stiffness.
- ☐ Lead-screw or gear pitch error from the component accuracy grade.
- ☐ Thermal growth of steel screws: about 11.5 µm per metre per K of temperature rise.
- ☐ Servo settling at standstill: allow ±1 to ±2 counts.
- ☐ Add repeatability terms separately from accuracy-only terms that can be mapped out.

Section 4 — Dynamics and inertia

- ☐ Peak speed for a triangular move profile = 2 × distance / move time.
- ☐ Reflected inertia = load inertia / ratio²; a carriage on a lead screw adds m × (lead in metres / 6.283)².
- ☐ Keep the reflected-to-motor inertia ratio below about 5:1 for responsive, well-damped moves.
- ☐ Acceleration torque = total inertia × angular acceleration; then add friction and process torque.
- ☐ Check RMS torque against the continuous rating and the highest torque against the peak rating.

Section 5 — Holding, safety and supplier data

- [ ] Fit a fail-safe brake on vertical axes and on axes that hold a process force.
- [ ] Use the brake to hold after an electrical stop, not as the main stopping device.
- [ ] Do not treat a self-locking worm gear as a safety brake.
- [ ] Define the homing routine and limit-switch positions.
- [ ] Record supply voltage and preferred command interface (0–5 V analogue, PWM, RS-485 Modbus or CAN).
- [ ] Attach the mounting interface, shaft details, envelope drawing and environment (ambient range, IP rating).

Encoder Resolution Quick Reference

Feedback	Counts per motor rev	Degrees per count at motor	Degrees per count at 30:1 output	mm per count, 10 mm lead direct
Hall sensors, 4 pole pairs	24	15	0.5	0.417
100 PPR encoder	400	0.9	0.03	0.025
250 PPR encoder	1,000	0.36	0.012	0.01
500 PPR encoder	2,000	0.18	0.006	0.005
1,000 PPR encoder	4,000	0.09	0.003	0.0025
2,500 PPR encoder	10,000	0.036	0.0012	0.001
12-bit absolute encoder	4,096	0.088	0.0029	0.0024

Resolution figures exclude backlash, compliance and lead error. Use the gearbox, screw and encoder data sheets for the final error budget.

Need help sizing or replacing a motor?
Globe Scott Motors has manufactured electric motors in Gujarat since 1985 and today builds PMDC, BLDC and geared motors, PMDC transaxles and railway signalling motors. Share your load, duty cycle and envelope at www.globescott.net/contact-us/

General engineering guidance only. Always follow the equipment manufacturer's instructions, applicable standards and site safety rules (isolation and lock-out/tag-out) before working on electrical or rotating equipment.